6.3.3 Models for Heat Baths and Langevin's Model for Brownian Motion

If the Ornstein-Uhlenbeck SDE models the momentum of particles undergoing collisions, it can be complemented by a corresponding differential equation for the position of the form

$$\mathrm{d}q = m^{-1} p \mathrm{d}t,$$

to provide a full description of the motion. This is the key idea in Langevin's model for Brownian motion [213].

Let us derive the Langevin dynamics model for the special case of a heavy particle (mass one) interacting with a large collection of light particles. We assume as in [138] that we are given a system of k coupled oscillators (a "heat bath") in contact with a distinguished particle. The Hamiltonian is

$$H = \frac{P^2}{2} + \sum_{i=1}^{k} \frac{p_i^2}{2\mu_i} + U(Q) + \frac{1}{2k} \sum_{i=1}^{k} (q_i - Q)^2$$

where Q and P represent the position and momentum of the distinguished particle and $\{q_i, p_i\}$ are the positions and momenta of the particles of the bath, μ_i giving the mass of the ith light particle.

The equations of motion for the combined system are

$$\dot{Q} = P, \tag{6.22}$$

$$\dot{P} = -U'(Q) - k^{-1} \sum_{i=1}^{k} (Q - q_i), \tag{6.23}$$

where, for $i = 1, 2, \ldots, k$,

$$\dot{q}_i = p_i/\mu_i, \tag{6.24}$$

$$\dot{p}_i = (Q - q_i)/k. \tag{6.25}$$

Because the system (6.24)–(6.25) is linear, we may solve it analytically for $\{q_i, p_i\}$; this yields:

$$q_i(t) = \cos(\Omega_i t) q_i(0) + \Omega_i^{-1} \sin(\Omega_i t) p_i(0)/\mu_i + k^{-1} \mu_i^{-1} \int_0^t \Omega_i^{-1} \sin(\Omega_i (t-s)) Q(s) \mathrm{d}s, \tag{6.26}$$

where $\Omega_i = k^{-1/2}\mu_i^{-1/2}$ is the frequency of the ith oscillator. Integrating by parts, we have

$$\int_0^t \Omega_i \sin(\Omega_i(t-s))Q(s)\mathrm{d}s = [\cos(\Omega_i(t-s))Q(s)]_0^t - \int_0^t \cos(\Omega_i(t-s))P(s)\mathrm{d}s$$

$$= Q(t) - \cos(\Omega_i t)Q(0) - \int_0^t \cos(\Omega_i(t-s))P(s)\mathrm{d}s.$$

which leads to

$$q_i(t) - Q(t) = \cos(\Omega_i t)q_i(0) + \Omega_i^{-1}\sin(\Omega_i t)p_i/\mu_i - \cos(\Omega_i t)Q(0) - \int_0^t \cos(\Omega_i(t-s))P(s)\mathrm{d}s.$$

Introducing this formula in (6.23), we obtain a reduced system

$$\dot{Q} = P, \tag{6.27}$$

$$\dot{P} = -U'(Q) - k^{-1}\sum_{i=1}^{k}(Q - q_i), \tag{6.28}$$

$$= -U'(Q) + f(t) - \int_0^t \psi(t-s)P(s)\mathrm{d}s, \tag{6.29}$$

where we define

$$f(t) = -k^{-1}\sum_{i=1}^{k}\left[\cos(\Omega_i t)q_i(0) + \Omega_i^{-1}\sin(\Omega_i t)p_i(0)/\mu_i - \cos(\Omega_i t)Q(0)\right],$$

$$\psi(t) = k^{-1}\sum_{i=1}^{k}\cos(\Omega_i(t)).$$

For large k and for many choices of the frequencies Ω_i and the initial conditions $\{q_i(0), p_i(0)\}$, function $f(t)$ will take on the character of a Wiener process; in fact the convergence can be shown in the weak sense. We may even take the initial conditions to be randomly distributed [13, 205]. The function $\psi(t)$ will typically exhibit an initial decay followed by rapid bounded fluctuation (see Fig. 6.4), with decay to zero in the limit of large k. Let $\{\Omega_i^{(k)} | i = 1, 2, \ldots, k\}$ represent the frequencies used for a given k. We may make various proposals for the $\Omega_i^{(k)}$,